RITUXIMAB

Infusion reactions — slow titration

Screen hepatitis B

CHEMICAL NAME Chimeric (mouse–human) IgG1κ monoclonal antibody against CD20
CLASS Monoclonal antibody — anti-CD20 (B-cell depleting)
TRADE NAMES MabThera · Rituxan · Truxima · Rixathon (biosimilars)

THERAPEUTIC USES / INDICATIONS

- B-cell non-Hodgkin lymphoma (R-CHOP), chronic lymphocytic leukaemia.
- Rheumatoid arthritis (with methotrexate), ANCA vasculitis, pemphigus vulgaris.
- Off-label: ITP, membranous nephropathy, MS (ocrelizumab is related), autoimmune haemolytic anaemia.

ADMINISTRATION

- **IV** infusion — first infusion started slowly (50 mg/h), increased by 50 mg/h every 30 min to max 400 mg/h; later infusions faster.
- Premedication (paracetamol, antihistamine) ± methylprednisolone (RA, vasculitis).
- SC formulation (lymphoma) after first IV dose.
- Do not give as IV push or bolus.

DOSAGES

- **Lymphoma:** 375 mg/m² per cycle (with chemotherapy) or weekly ×4.
- **RA:** 1,000 mg IV on days 1 and 15; repeat course ≥6 months.
- **Vasculitis:** 375 mg/m² weekly ×4, or 1 g ×2.

PHARMACOKINETICS

- B-cell depletion within days; recovery 6–12 months.
- **t½:** ~3 weeks (variable).

PHARMACODYNAMICS

- Binds **CD20** on B lymphocytes → cell lysis via complement-dependent cytotoxicity, ADCC and apoptosis → B-cell depletion.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Infusion-related reactions: fever, chills, rigors, hypotension, bronchospasm (mostly first infusion).
- Infections (respiratory).
- Fatigue, nausea.

ADVERSE EFFECTS (SERIOUS)

- Hepatitis B reactivation (fulminant hepatitis, death).
- Progressive multifocal leukoencephalopathy (PML).
- Tumour lysis syndrome (high tumour burden); severe mucocutaneous reactions.
- Hypogammaglobulinaemia; late neutropenia; cardiac arrhythmias.

CONTRAINDICATIONS & CAUTIONS

- Active severe infections; severely immunocompromised.
- Severe heart failure (NYHA IV) or uncontrolled cardiac disease (RA indication).
- Live vaccines.

KEY INTERACTIONS

- **Live vaccines:** contraindicated (and poor vaccine response for months).
- **Antihypertensives:** consider withholding 12 h before infusion (hypotension).

NURSING MANAGEMENT

- Screen hepatitis B (HBsAg, anti-HBc) before first dose.
- Vital signs every 15–30 min during infusion; titrate per protocol; stop for reactions.
- Monitor for tumour lysis (lymphoma) — hydration, allopurinol.
- Check immunoglobulins and infection history before repeat courses.

PATIENT EDUCATION

- Report fever, chills, itching, breathlessness during the infusion.
- Report infections, confusion or vision/balance problems (PML) promptly.
- Avoid live vaccines; get non-live vaccines ideally before treatment.
- Consider withholding BP tablets on infusion day if advised.

KEY POINTS

- Anti-CD20 B-cell depleting antibody.
- Slow first infusion; premedicate.
- Lymphoma 375 mg/m²; RA 1 g × 2.
- Hepatitis B reactivation, PML.

SOURCES CHECKED **[916]** MabThera 100 mg Concentrate for Solution for Infusion (UK SmPC) **[917]** Rituxan, IV (US label)
Full references p. 432–457